

UTKARSHA GHOSH

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EDUCATIONAL QUALIFICATIONS

- Pursuing B.tech in Information technology (2023-2027) from Institute of Engineering and Management , Saltlake , Kolkata
- Class XII, Central Board of Secondary Education , 2023, Computer Science Stream, South Point high School, Ballygunge , Kolkata with 84.8%.
- Class X , Central Board of Secondary Education , 2021, South Point high School, Ballygunge , Kolkata with 94.6%.

PROFESSIONAL SUMMARY

I am a third-year B.Tech student in Information Technology at the Institute of Engineering and Management. Proficient in Python, React, and cloud technologies, I have developed AI-driven platforms like AuraSync and published research on hierarchical opinion classification. I am a dedicated problem-solver focused on applying technical expertise through innovation and team collaboration.

TECHNICAL SKILLS

Programming Languages: Python, C, HTML , Java , Powershell

Libraries and Frameworks : React , Java Collections Framework , Python libraries , Pandas , Tensorflow , numpy , scikit-learn

Database : Firebase Realtime Database / Cloud Firestore, Supabase, MySQL.

Technologies: VS Code, Jupyter Notebook , Google Colab , Powershell , Git , Github , LaTeX

PROJECTS

- **AuraSync (React , TypeScript , Supabase , Spotify Web API , Face-api.js)**
Built an AI-powered music recommendation platform that uses facial recognition to detect user mood and syncs with Spotify.
Implemented real-time emotion detection using **face-api.js** and designed a responsive, accessible UI with **shadcn/ui** and **Tailwind CSS**.
- **Soulscribe (React, TypeScript, Supabase, Gemini 1.5 Flash)**
Built an AI-driven wellness platform
Implemented **Gemini 1.5 Flash** for real-time empathetic dialogue, including safety guardrails for crisis intervention.
Utilized **React Query** for efficient state management and data fetching from a **Supabase** PostgreSQL backend.
- **Windows System Monitoring Tool (Python, PowerShell, Windows Event Logs , Pandas)**
Built a log analysis tool that automates system health monitoring via **PowerShell** Get-WinEvent integration. Implemented persistent error tracking and data deduplication using **Python** and **JSON**. Automated the generation of multi-sheet Excel reports for system audit compliance using **Pandas**.

INTERNSHIPS

**THE NATIONAL SMALL INDUSTRIES CORPORATION LTD. TECHNICAL SERVICES CENTRE
[SEPTEMBER 12 , 2024 – SEPTEMBER 26,2024] 2 WEEKS**

- Completed the internship program in the area of Snowflake database in Cloud

IT Intern | Emami Agrotech Limited | [June 2025 – July 2025] 2 MONTHS

- Developed and deployed a centralized system for monitoring system logs from over 40 Windows servers, enabling proactive issue identification.
- Implemented and configured CMDBuild, an open-source IT asset management system, to replace a manual, spreadsheet-based tracking process

PUBLICATIONS

- **Hierarchical Opinion Classification with Large Language Models** 2025
Submitted and Accepted in FIRE 2025
Independent Research with Co-authors: Alok Mishra and Shuvam Banerji Seal

Proposed a parameter-efficient fine-tuning approach for hierarchical opinion classification using the Gemma-1B model, employing a lightweight two-layer classification head, selective transformer block updates, and an 8-class flat reformulation with class-weighted loss to enhance minority-class performance and structural sentiment understanding.

COURSES

• NPTEL

- The Joy of Computing using Python (IIT Madras) – July-October 2023
- Google Cloud Computing Foundations (IIT Kharagpur) – August – October 2023
- Introduction to Algorithms and Analysis (IIT Kharagpur) – July – October 2023
- Cyber Security and Privacy (IIT Madras) – July – October 2024

• COURSERA

- Advanced System Security Topics (University of Colorado)
- Probability & Statistics for Machine Learning & Data Science (Deeplearning.AI)
- Information Theory (The Chinese University of Hong Kong)

• LINKEDIN LEARNING

- Python Object Oriented programming

• Online Summer School on Usefulness of Remote Sensing & GIS for Environmental Studies Conducted by Indian Institute of Remote Sensing , ISRO Dehradun

ACHIEVEMENTS

- Awarded a Gold Medal of Excellence for performance excellence at the school level in SOF – National Science Olympiad (2021-2022)
- Qualified for Level-2 in Techtriad Hackathon in KSHITIJ 2025

I hereby declare that the aforesaid information is true to the best of my knowledge.

